Tutorial 3

CS3230 · Divide and conquer

Demo course material, written for OpenPapr

Questions

- Q1. Solve $T(n) = 4T(n/2) + n$ with the master theorem
- Q2. Solve $T(n) = T(n/3) + T(2n/3) + n$ with a recursion tree
- Q3. Count the inversions in an array in $O(n \log n)$

Hints

- Q1: compare n with n^(log_2 4)
- Q2: the tree is lopsided; bound its depth
- Q3: count while merging